

The Information Paradox Resolved:

0/0 of Black Hole Information

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The Removable Singularity Framework

Abstract

For 45 years, Hawking's calculation showed black holes destroy information, violating quantum mechanics (unitarity). This is the original 0/0: in formation at the horizon is neither destroyed nor preserved. Page (1993) showed unitarity requires a specific entropy curve (the Page curve). In 2019-2020, Penington, Almheiri-Engelhardt, and Wall derived the ISLAND FORMULA: $S_{\text{rad}} = \min(\text{Area}(\text{island})/4G_N + S_{\text{semiclassical}})$. Replica wormholes reproduce the Page curve EXACTLY. Information IS conserved through entanglement. The 0/0 IS resolved.

1. Hawking (1975)

Black holes radiate (Hawking radiation). The radiation is purely thermal: it contains NO information. As the black hole evaporates, information appears destroyed. This violates unitarity (quantum mechanics). This is the original information 0/0.

2. Page Curve (1993)

Unitarity requires information conservation. The entropy of radiation must follow a SPECIFIC curve: increase to $S_{\text{BH}}/2$ at Page time, then DECREASE back to zero as the black hole fully evaporates. At $t=t_{\text{evap}}$, $S_{\text{rad}} = 0$ (ALL information escaped).

3. Island Formula (2020)

$$S_{\text{rad}}(R) = \min(\text{Area}(\text{island}) / 4G_N + S_{\text{semiclassical}})$$

The island is the entanglement wedge INSIDE the black hole. The replica trick computes $\text{Tr}(\rho^n)$ directly. Before Page time: Hawking result (no wormhole). After Page time: replica wormholes CONNECT the copies, giving the island contribution. The result reproduces the Page curve EXACTLY.

4. The 0/0 Is Resolved

The horizon 0/0 (information neither destroyed nor preserved) is RESOLVED by quantum gravity:

- Information IS conserved (unitarity)
- It escapes through ENTANGLEMENT, not radiation
- The island: interior becomes part of radiation wedge
- Replica wormholes: the mechanism

The 0/0 is not destroyed or preserved independently -- it is SHARED through entanglement.

5. Connections

System	Prior Chapter	Connection
Quantum gravity	Ch.51 QG	The mechanism
Holographic	Ch.47 Holographic	Ryu-Takayanagi
Black holes	Ch.32 Black holes	Original setting
Measurement	Ch.49 Measurement	Entanglement
RMT	Ch.44 RMT	SYK model

6. Conclusion

The information paradox IS RESOLVED. Hawking (1975) predicted information destruction; Page (1993) required conservation; islands (2020) proved it. The island formula $S = \min(\text{Area}/4G_N + S_{\text{semiclassical}})$ reproduced the Page curve EXACTLY via replica wormholes. Information IS conserved through entanglement. This is the first PROOF that quantum gravity works, resolving the 0/0 of black hole information.

References

- [1] Hawking (1975), particle creation by black holes
- [2] Page (1993), information in black hole radiation
- [3] Penington (2019), entanglement wedge reconstruction
- [4] Almheiri, Engelhardt, Marolf, Maxfield (2019), entropy of radiation
- [5] Almheiri, Hartman, Maldacena, Shaghoulian, Tajdini (2020), replica wormholes
- [6] Puno, "The Removable Singularity" (2026)